ASSIGNMENT 3

1. List the various CATEGORIES and COUNT OF BOOKS in each category.
2. List the BOOK_NO and the NUMBER OF TIMES the Book is issued in the descending order of COUNT.
3. Display the MEMBER ID and the NO OF BOOKS for each member that has issued more than 1 book.
4. Display the MEMBER ID, BOOK NO and NO OF TIMES the same book is issued by the MEMBER in the descending order of count.
5. Select all information of Book with 2nd Maximum Cost.
6. Copy the MEMBER, BOOKS and ISSUE tables into MEMBER_COPY, BOOKS_COPY, ISSUE_COPY respectively.
7. Create a table named QUERY1 with two attributes as BOOK_NO, ISSUE_DATE and copy all information about BOOK_NO and ISSUE_DATE from ISSUE table to QUERY1 table.
8. Create and store BOOK_NAME, AUTHOR_NAME and COST to QUERY2 table where cost is greater than 300.
9. Insert BOOK_NAME, AUTHOR_NAME and COST to QUERY2 table where cost is less than or equal to 300.
10. ADD an attribute name AVAILABLE with data type NUMBER (5) in the BOOKS_COPY table and fill up the new attribute with data.
11. Change the Data type of CATEGORY attribute to VARCHAR2 (15) in BOOKS_COPY table.

ASSIGNMENT 2

1. Retrieve the Name of Book and Cost which has Maximum cost.
   SELECT Book_Name, Cost FROM BOOKS WHERE COST=(SELECT MAX(COST) FROM BOOKS)

2. Calculate the Minimum cost, Average cost and Total cost value in BOOKS table and Rename the resulting attributes.
   SELECT MIN(COST) AS MINIMUM_COST, AVG(COST) AS AVERAGE_COST, SUM(COST) AS TOTAL_COST FROM BOOKS

3. Retrieve the Name and ID of Members who have issued book between 26th January 2011 and 14th April 2011.
   SELECT MEMBER.MEMBER_ID, MEMBER_NAME, ISSUE_DATE FROM MEMBER, ISSUE WHERE MEMBER.MEMBER_ID=ISSUE.MEMBER_ID AND ISSUE_DATE BETWEEN '26-JAN-2011' AND '14-APR-2011'

4. Retrieve Book Name, Author Name and Category whose Category is not ‘OTHERS’.
   SELECT BOOK_NAME, AUTHOR_NAME, CATEGORY FROM BOOKS WHERE NOT CATEGORY='Others'

5. Retrieve the Book name and Author Name where 5th letter of the Author name is ’t’.
   SELECT BOOK_NAME, AUTHOR_NAME FROM BOOKS WHERE AUTHOR_NAME LIKE '____t%'

6. How many Books are available whose Cost is greater than 350.
   SELECT COUNT(BOOK_NO) FROM BOOKS WHERE COST>350

7. How many different Authors name are available in BOOKS table.
   SELECT COUNT(*) FROM (SELECT DISTINCT AUTHOR_NAME FROM BOOKS)

8. Calculate the following Numeric functions:
   a. What is the absolute value of -167.
      SELECT ABS(-167) FROM DUAL

   b. Calculate 8^6.
      SELECT POWER(8,6) FROM DUAL

   c. Round up to 2 decimal points (134.56789).
      SELECT ROUND(134.56789,2) FROM DUAL

   d. What is the square root of 144
      SELECT SQRT(144) FROM DUAL

   e. Floor and Ceil value of 13.15
      SELECT FLOOR(13.15), CEIL(13.15) FROM DUAL

9. Extract Year, Month, Day from System Date.
SELECT EXTRACT(YEAR FROM SYSDATE),EXTRACT(MONTH FROM SYSDATE),EXTRACT(DAY FROM SYSDATE) FROM DUAL

10. What is the greatest value between 4, 5 and 17.
SELECT GREATEST(4,7,17) FROM DUAL

11. What is the Least value between '4', '5' and '17' and Express why resulting value of last two queries are same.
SELECT LEAST(4,7,17) FROM DUAL

SELECT LEAST('4','7','17') FROM DUAL

LEAST('4','7','17')
17

12. Extract 4 letters from 3th position of this word 'INFOSYS'.
SELECT SUBSTR('INFOSYS',3,4) FROM DUAL

13. What is the ASCII value of 'a' and 'S'.
SELECT ASCII('a'), ASCII('S') FROM DUAL

14. What is Length of this word 'INFOSYS' AND change 'S' with 'T'.
SELECT LENGTH('INFOSYS'), REPLACE('INFOSYS','S','T') FROM DUAL

15. Retrieve the Names and Address of the Members who belong to Kolkata.
SELECT MEMBER_NAME, MEMBER_ADDRESS FROM MEMBER WHERE MEMBER_ADDRESS='Kolkata'

16. Retrieve the Name of Books, where Cost prices are between 300 and 500.
SELECT BOOK_NAME,COST FROM BOOKS WHERE COST BETWEEN 300 AND 500
or
SELECT BOOK_NAME,COST FROM BOOKS WHERE COST>=300 AND COST<=500

17. List the Name of the Members whose Membership type is “HALF YEARLY”.
SELECT MEMBER_NAME FROM MEMBER WHERE MEMBERSHIP_TYPE='Half Yearly'

18. List the Name of the Members who open their account in the year of 2011.
SELECT MEMBER_NAME FROM MEMBER WHERE EXTRACT(YEAR FROM ACC_OPEN_DATE)=2011
or
SELECT MEMBER_NAME FROM MEMBER WHERE ACC_OPEN_DATE BETWEEN '01-JAN-11' AND '31-DEC-11'

19. Retrieve the Penalty Amount of the Members who has taken the book “LET US C”.

SELECT PENALTY_AMOUNT FROM MEMBER WHERE MEMBER_ID IN((SELECT MEMBER_ID FROM ISSUE WHERE ISSUE.BOOK_NO=(SELECT BOOK_NO FROM BOOKS WHERE BOOK_NAME='Let us C')))

20. Retrieve the no of Max books allowed to a Member, who has issued books on January.

SELECT MAX_BOOKS_ALLOWED FROM MEMBER WHERE MEMBER_ID IN (SELECT MEMBER_ID FROM ISSUE WHERE EXTRACT (MONTH FROM ISSUE_DATE)=1)

21. Give the Names of the Members who have not issued any books.

SELECT MEMBER_ID, MEMBER_NAME FROM MEMBER WHERE MEMBER_ID IN (SELECT MEMBER_ID FROM MEMBER MINUS SELECT MEMBER_ID FROM ISSUE)

22. Give the Name and Category of the books whose cost is not recorded.

SELECT BOOK_NAME, CATEGORY FROM BOOKS WHERE COST IS NULL

23. List all the books that are written by Author ‘’Loni’’ and has Price less than 600.

SELECT BOOK_NAME FROM BOOKS WHERE AUTHOR_NAME='Loni' and cost<600

24. List the Issue details for the Books that are not returned yet.

SELECT * FROM ISSUE WHERE RETURN_DATE IS NULL

25. List all the Books that belong to any one of the following categories: Science, Database.

SELECT BOOK_NAME, CATEGORY FROM BOOKS WHERE CATEGORY IN('Science','Database')

26. List all the Members in the descending order of Penalty due on them.

SELECT * FROM MEMBER ORDER BY(PENALTY_AMOUNT) DESC

27. List all the Books in ascending order of category and descending order of price.

SELECT * FROM BOOKS ORDER BY CATEGORY ASC , COST DESC

28. List all the Books that contain word ‘SQL’ in the Name of the Book.

SELECT BOOK_NAME FROM BOOKS WHERE BOOK_NAME LIKE '%SQL%'

29. List all the Members whose Name starts with S.

SELECT MEMBER_NAME FROM MEMBER WHERE MEMBER_NAME LIKE 'S%'

30. List all the Members whose Name starts with S or A and contains letter T in it.

SELECT MEMBER_NAME FROM MEMBER WHERE MEMBER_NAME LIKE 'S%t%' OR MEMBER_NAME LIKE 'A%t%'

31. List the Entire Book name in INIT CAP and Author Name in UPPER case in the descending order of the Book Name.

SELECT INITCAP(BOOK_NAME), UPPER(AUTHOR_NAME) FROM BOOKS ORDER BY BOOK_NAME DESC

32. List the data in the book table with category data displayed as ‘D’ for Database, ‘S’ for Science, ‘R’ for RDBMS and ‘O’ for all the others.

SELECT BOOK_NO, BOOK_NAME, AUTHOR_NAME, COST, SUBSTR(CATEGORY,1,1) FROM BOOKS

33. List all the Members that became the Member in the year 2011.

SELECT MEMBER_NAME, ACC_OPEN_DATE FROM MEMBER WHERE EXTRACT(YEAR FROM ACC_OPEN_DATE)=2011